

Divya D Kulkarni

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📍 Boston, MA, USA · Open to relocation

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Experience

- Feb 2024 – Present ■ **Postdoctoral Research Fellow – [Deep Neurophenotyping Lab](#), Massachusetts General Hospital & Harvard Medical School**
- **In-clinic ([Neurobooth](#), multimodal):** Lead research on video-based gaze estimation that predicts eye rotation in degrees directly from smartphone video, rather than screen-coordinate gaze, achieving sub-3° error against a 1000 Hz research-grade eye tracker (EyeLink) for detecting cerebellar oculomotor abnormalities; evaluated with cross-validation designed to prevent temporal data leakage.
 - Co-develop the [Neurobooth](#) multimodal acquisition platform: synchronized recording across 18 sensors (cameras, microphones, body-worn IMUs, eye tracker) via Lab Streaming Layer, aligning streams with heterogeneous sample rates (variable 180–240 Hz video to 1000 Hz IMU and eyetracking) for multimodal learning across video, speech, eye tracking, and motion.
 - Training a single task-general vision model for eye tracking across the Neurobooth battery of oculomotor and cognitive tasks and across disease populations, on smartphone video with distributed multi-GPU training (PyTorch DDP).
 - **At-home (remote wearable studies):** Build time-series models on continuous triaxial accelerometer data from week-long at-home wearable recordings to model longitudinal disease progression in Ataxia-Telangiectasia and Friedreich's Ataxia (ICAR 2024; invited talk, A-T Clinical Research Conference 2025).
 - All models are built on two prospective clinical data collections, not curated benchmarks, requiring extensive cleaning, missing-data handling, and handling of severe class imbalance and out-of-distribution shifts.
- Oct 2022 – Oct 2023 ■ **Machine Learning Consultant – Eli Lilly and Company, India**
- Built computer vision defect-detection models to identify cracked drug vials in pharmaceutical manufacturing; subsequently deployed to production.
 - Developed time-series anomaly detection on equipment sensor data to localize failing components in assembly lines.
 - Designed segmentation models to quantify drug absorption in stained histopathology slides, where non-uniform stain distribution defeated standard segmentation approaches.
- Feb 2022 – Sept 2022 ■ **Machine Learning Intern – Eli Lilly and Company, India**
- Developed vision-based obstacle avoidance for an autonomous ground vehicle equipped with only a single front distance sensor and an onboard camera, combining deep-learning object detection with classical computer vision to localize a projected light marker for navigation; subsequently deployed to production. Converted to a full-time role.
- Aug 2014 – Aug 2015 ■ **Member of Technical Staff – MetricStream, Bangalore, India:** internal tools for a governance, risk, and compliance software platform.

Education

- 2016 – 2023 ■ **Ph.D., Computer Science and Engineering, Indian Institute of Technology (IIT) Guwahati, India**
Thesis: *Immuno-Inspired Embodied Lifelong Learning in Robots* (submitted Sept 2022; defended May 2023).
- 2010 – 2014 ■ **Bachelor of Engineering, Computer Science and Engineering, SDM CET, Dharwad, India**

Awards

- 2021 ■ **Best Student Paper Award**, IEEE Congress on Evolutionary Computation (CEC), Kraków, Poland – *An Immuno-Inspired Transfer Learning Paradigm*
- 2021, 2022 ■ **Google Research (India) Symposium** – selected twice as one of the top PhD students in the Asia-Pacific region

Selected Publications

- 1 Oubre, B., Yang, F., Luddy, A. C., Manohar, R., Soja, N. N., Stephen, C. D., Schmahmann, J. D., **Kulkarni, D. D.**, White, L., Patel, S., & Gupta, A. S. (2026). A digital measure of eye movements during reading sensitively captures oculomotor and speech dysfunction, early changes, and disease progression in ataxias. *Annals of Neurology*, 99(1), 84–95. <https://doi.org/10.1002/ana.78039>
- 2 **Kulkarni, D. D.**, & Nair, S. B. (2023). Transfer learning for embodied neuroevolution. *Neuroevolution at Work Workshop, Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO)*.
- 3 **Kulkarni, D. D.**, Semwal, T., & Nair, S. B. (2022). Immuno-inspired management of halls of fame for embodied evolution. *Swarm and Evolutionary Computation*, 70, 101054. <https://doi.org/10.1016/j.swevo.2022.101054>
- 4 **Kulkarni, D. D.**, & Nair, S. B. (2021). An immuno-inspired transfer learning paradigm. *IEEE Congress on Evolutionary Computation (CEC) (Winner - Best Student Paper Award)*.
- 5 Agrawal, A., **Kulkarni, D. D.**, Semwal, T., & Nair, S. B. (2020). On decentralizing federated learning. *IEEE Conference on Systems, Man and Cybernetics (SMC)*.
- 6 **Kulkarni, D. D.**, & Nair, S. B. (2020). Mutational puissance assisted neuroevolution. *Neuroevolution at Work Workshop, Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO)*, 1841–1848. <https://doi.org/10.1145/3377929.3398149>
- 7 Semwal, T., **Kulkarni, D. D.**, & Nair, S. B. (2018). On an immuno-inspired distributed, embodied action-evolution cum selection algorithm. *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO), Kyoto, Japan*, 141–148.

Posters & Invited Talks

- 1 **Kulkarni, D. D.**, et al. (2026). Automatic video gaze estimation for detection of cerebellar oculomotor abnormalities in ataxias [Poster, International Congress for Ataxia Research (ICAR). Full manuscript in preparation].
- 2 **Kulkarni, D. D.** (2025). Disease progression in Ataxia-Telangiectasia – at-home wearables study [Invited Speaker, Ataxia-Telangiectasia Clinical Research Conference, Loughborough, United Kingdom].
- 3 **Kulkarni, D. D.**, et al. (2024). Movement patterns from at-home wearables differentiate patients with Ataxia-Telangiectasia (A-T) and Friedreich’s Ataxia (FA) from healthy controls [Poster, International Congress for Ataxia Research (ICAR), London, United Kingdom. Full manuscript in preparation].
- 4 **Kulkarni, D. D.** (2023). Transfer neurons not layers [Women in Machine Learning (WiML) Workshop, NeurIPS 2023, New Orleans, USA].

Technical Proficiencies

Languages	■ Python, C++, SQL, Prolog
ML Frameworks	■ PyTorch, HuggingFace Transformers, scikit-learn, OpenCV
ML Areas	■ Computer Vision, Multimodal Learning, Time-Series Modeling, Transformer & Attention Architectures, Transfer Learning, Neuroevolution
Tools & Infra	■ Git, Linux, Docker, AWS, Lab Streaming Layer (LSL), ROS, TensorBoard, $\LaTeX$

Software Projects

2020 – Present	■ Creator – TarPy (Multi-Agent Distributed Learning Framework) • Designed and built TarPy, a Python framework for multi-agent distributed learning research, with end-to-end data ingestion, distributed model training, and evaluation pipelines. <i>Open-source release in preparation.</i>
2017 – 2022	■ Core Developer – <u>Tartarus</u> , an open-source multi-agent platform in SWI-Prolog, IIT Guwahati

Academic Service & Leadership

Reviewer	■ CHIL 2026; ML4H 2024–2025; IJCNN 2025–2026; AI for Medical 3D Vision (AI4M3D) Workshop, ECCV 2026; Women in Machine Learning (WiML) Workshop, NeurIPS 2026
2021	■ Session Chair , “Evolved Neural Networks – II”, IEEE Congress on Evolutionary Computation (CEC)
Mentoring	■ Mentored 6 thesis students (1 undergraduate, 5 master’s); Head TA, Mobile Robotics (2017–2021)